

## Western Blot Protocol

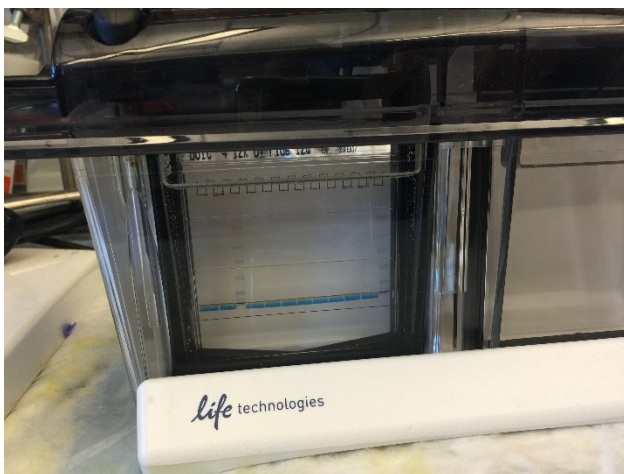
- **Updated 6/8/22**

### Sample Preparation

1. *Sample Preparation*
  - a. Do BCA to all samples and adjust to equal protein concentration
  - b. 1mg/mL protein is a good range for western blot
2. Loading Buffer (in 4C fridge) “NuPAGE (4X)”, small bottle with blue solution
  - a. Sample : Loading Buffer = 3 : 1
3. Sample amount: 15~20uL/channel
  - a. If two channels, 40uL in total □ 45uL sample +15uL NuPAGE
  - b. *If proteins accumulate on top (at the end of first run), next time add BME into sample, 1:20. Do this in fume hood.*
4. turn on water heater to 70C. Heat samples at 70C for 10min (**Longer time up to 1hr**), meanwhile:

### First Run

5. Find running buffer (475mL milliQ water + 25mL 20X Running buffer (serological))
6. Take running gel board from 4C fridge, Invitrogen box.
  - a. Open pouch over sink, pour water out
  - b. Rinse with pure water, remove “comb” on top carefully
  - c. **Take off the white insulation strip**
  - d. Put gel board into cassette, lock
  - e. Add running buffer **Add running buffer into the wells carefully**
7. Take samples from water bath, carefully add samples to each well, should see purple squares. Ladder solution—“SeeBlue Plus 2 Prestained”, 5uL/channel
  - a. Sample locations should NOT be symmetrical (transfer step may be confusing)
8. 165V, 45min.



### Transfer Gel

9. Take out cassette, rinse with pure water, keep the gel wet

10. Open cassette, put gel in water, cut off bottom and top **Rinse gel with water a few times**
11. 2 Sponges, 2 Filter paper, 1 membrane, all cut into same size as sponge
12. Submerge EVERYTHING in 400mL Transfer Buffer **Remove sponge bubble by pressing down**
13. Assemble following protocol, use roller to remove bubbles
14. Put cassette into gel system, pour 100% transfer buffer onto cassette, pour water to the side
15. Quick Blot 15V 30min



### **BSA Blocking**

16. Make 5% milk, 2g/40mL x2 (depends on number of membrane, 5mL/section)
17. Add BSA 1~2% (0.4g or 0.8g/40mL) (**optional** step for high background)
18. Dissolve in PBST (0.05% Tween 20)
19. Sonicate/**Vortex** milk for 2min
20. Take out membrane, rinse with water
21. Submerge in Ponceau to see ladders (**Seeing ladder, pour ponceau back and water rinse**)
22. Cut membrane into 2~3 channel wide strips, and cut into smaller squares based on blue ladder
  - a. Remember sample groups, cut corner on top left to remember
23. Rinse membrane **with water to remove ponceau as much as possible**
24. Rinse incubation box with PBST
25. Add 4mL milk solution to each compartment, submerge membrane within
26. Incubate at RT for 1~2 hours to block non-specific binding, change milk after 20min

### **Primary Antibody**

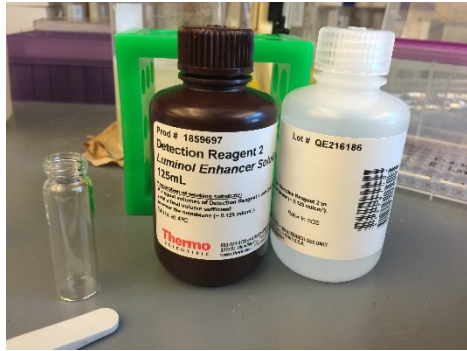
27. Prepare antibody dilution in milk solution, usually 1:1000 dilution is good
  - a. One antibody per box section to avoid cross reaction
28. Rinse membrane and box
29. Incubate at 4C overnight

### **Secondary Antibody**

30. Repeat 29, 30, 31 for secondary antibody solutions, **secondary at least 2hrs, 1:1000**
31. **3 x 5min before secondary; 3 x 7-10min washes after secondary**

### **Imaging**

32. Rinse membrane 3 times with PBST **water rinse PBST off (PBST affects substrate)**



33. Prepare developing solution, two reagents 1.5mL + 1.5mL  $\square$  3mL.
  - a. Use solution to cover the membrane (**at least 500uL each**)
  - b. Incubate for 2-3 minutes
  - c. Take solutions away
34. Put membrane in the cassette, take film, fourth floor, dark room.



35. Close door tightly, put one film into the cassette, count time, 3min-3hrs
36. Could do multiple films to see darker colors
37. Roll exposed film into machine, from left corner.